Chapter 4
Print ISBN: 978-93-5547-445-2, eBook ISBN: 978-93-5547-451-3

Dasatinib and Quercetin: Short-term Simultaneous Administration Yields Senolytic Effect in Humans

Tkemaladze Jaba [a*]

DOI: 10.9734/bpi/idmmr/v2/15155D

ABSTRACT

Background: The more divisions of cells, the more senescent cells. Senescent cells are one of the most toxic cells in human and animal organisms. When their amount reaches great quantities, it leads to strong manifestation of ageing signs. According to the explanation of the citogenetic centriolar theory of ageing during the division process, the cell's generation reaches ultimate morphogenetic status. As a result, programmed cell death (apoptosis) must be activated, otherwise if the cell does not kill itself it switches to a senescent state.
Purpose: Senescent cells cause age-related pathologies. Theoretically, partial elimination of senescent cells must lead to improved life capacity. This hypothesis was confirmed on Ercc1 -/Δ mice. Periodic simultaneous administration of dasatinib and quercetin in Ercc1 -/Δ mice led to extension of their healthspan, delaying age-related symptoms and pathologies. We decided to test the senolytic effect of dasatinib and quercetin combination on humans.
Study Design: With this purpose the clinical trial was undertaken on 64 male volunteers over the age of 36. To put things into perspective our volunteers were classified into 4 groups with 16 people in each. D+Q group orally administered 50mg of dasatinib with 500 mg of quercetin once a day within 5 days.
Methods: To register and to assess changes stemming from the drug compounds the whole trial was complemented with full medical screening and stair ascending test.
Results: As a result 50mg of dasatinib along with 500 mg of quercetin demonstrated obvious senolytic effect, it was justified by improved results of stair ascending test and tranquilized state of systolic blood pressure. Such dosage combination of these two compounds is highly probable to be harmless.

Keywords: Senolytic; senescent cells; dasatinib; quercetin; ageing; lifespan; cell divisions; centriolar; cytogenetic.

1. INTRODUCTION

It is proven that senescent cells are the chief source of intoxication to multicellular animal/human organisms [1-2]. Mechanism of their origin is hitherto indistinct. Theoretical basis for experiment of partial elimination of senescent cells from an organism has been *the Cytogenetic centriolar theory of ageing* [3-4].

The destruction and dissipation of centrioles during prophase of the first meiotic division [5] perhaps testifies that the cell begins the process of acquisition of totipotency- "null" morphogenetic status.

As sperm and oocyte undergo fusion, the resulting totipotent zygote divides and gives rise to the first generation - 2 totipotent cells. These cells also divide and produce the next generation of totipotent cells, and thus it continues until the formation of de novo centrioles at a certain generation [6-7]. From each separate cell (without centrioles) can develop monozygotic (MZ) twins.

[a] *International Longevity Centre- 0131, King Demetre Tavdadebuli 81, Tbilisi, Georgia.*
**Corresponding author: E-mail: djabbat@gmail.com, jaba@longevity.ge;*

From the formation of centrioles cells of that generation lose their totipotency [8] and turn to the path of irreversible differentiation - they acquire a certain morphogenetic status that changes in the next-generation lineage [9]. Alteration of morphogenetic status signifies that one gene network has been turned off, while the other turned on [10]. The number of cell generations, starting from zygote, is limited - Hayflick limit [11].

Programmed cell death becomes activated in the cells of the last generation.

According to the Cytogenetic centriolar theory of ageing it is possible that:

1. The structure of the division ‘counter’ of somatic cells is physically connected with centrioles. These structures consist of a chain of different small molecule links of RNA/proteins. One structure is connected with one centriole, whereas the other is connected with another centriole, each with a different set of small molecules [12].
2. After the fertilization, prior to de novo formation of a pair of centrioles, a pair of the structures of divisions ‘counter’ is generated de novo on the basis of nuclear/mitochondrial DNA Matrix (Fig. 1).

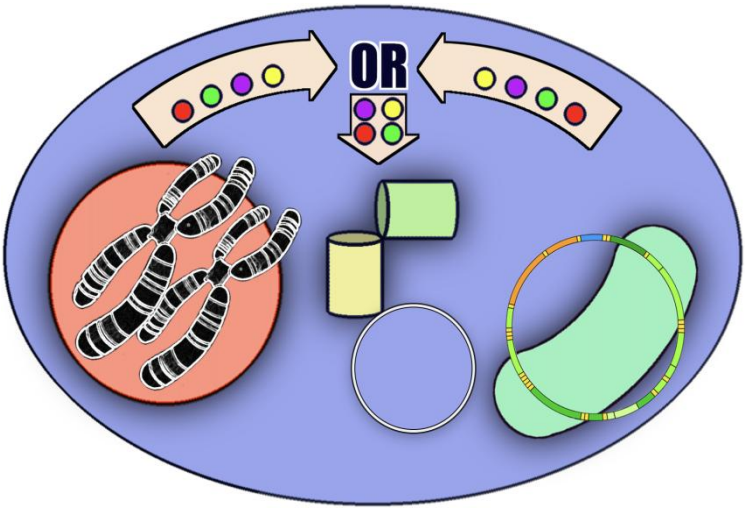

Fig. 1.

Each centriole of centrosomes encapsulates various molecular structures of divisions ‘counter’. When one looks at mutually perpendicular centrioles stationed with their angle downward, the left centriole encloses the left structure of divisions ‘counter’ (L-structure), whereas the right centriole encloses the right structure of divisions ‘counter’ (D-structure).

3. After mitosis, prior to reduplication of centrioles, reduplication of divisions 'counter’ structures takes place in daughter cells. Due to this, the mother centriole with L-structure bears daughter centriole with D-structure, and mother centriole with D-structure bears daughter centriole with L-structure.
4. After certain mitosis, at the very beginning of the reduplication process of structures of divisions ‘counter’, one small molecule of RNA/protein unchains from the mother ‘counter’ structure. In one daughter cell it unchains from the L-structure of mother centriole (nothing unchains from the D-structure of the second centriole), and in the other daughter cell one small molecule unchains from the D-structure of the mother centriole (nothing unchains from the L-structure of the second centriole). These are two various molecules in two different daughter cells (Fig. 2).
5. These small molecules of RNA/protein enter from cytoplasm into the karyoplasm of the cell nucleus. Then these small molecules of RNA/protein transcribe into small molecules of DNA. Afterwards these small molecules of DNA insert into the structure of nuclear DNA - editing of nuclear DNA takes place. Editing, modification of nuclear DNA structures causes turning-off of one gene network and turning-on of the other - morphogenetic status alters. Thus

morphogenetic status can be identified not only by gene network turned-on in nuclear DNA, but also by the presence of one or another small molecule of RNA/protein in L and D structures of divisions ‘counter’. Unless small molecules in one of (L or D) structures of divisions ‘counter’ are exhausted, morphogenetic status alters in the same generation in both daughter cells.

6. As removal of the small molecule of RNA/protein from the structure of divisions ‘counter’ occurs asymmetrically (only from the structure of the mother centriole), one of the structures of divisions ‘counter’ shortens quicker than the other. Thus, a cytogenetic vector emerges in a certain generation that can produce an infinite number of generations in which one of the daughter cells always has the same structure of divisions ‘counter’ as does the mother cell with the same unchanged morphogenetic status. The amount of such cytogenetic vectors can be the same as small molecules of RNA/protein present in both structures of divisions ‘counter’.
7. Meanwhile a small molecule unchains in the second daughter cell of such cytogenetic vectors, from the mother centriole’s structure of divisions ‘counter’, as a consequence morphogenetic status alters. Thus only asymmetrical divisions are possible in such cytogenetic vectors - morphogenetic status of only one daughter cell alters, and morphogenetic status of the other daughter cell remains identical to that of mother’s.
8. It is most likely that free radicals/nitrogen oxide (NO) cause the unchaining of small molecules of RNA/protein from the structure of divisions ‘counter’. Probably this takes place not in every mitosis, that is the ‘counter’ does not keep count of each division. The probability of such an occasion most likely depends on the sizes, geometries of cells, location of centriole (it is not always in the center of the cell) and also on the concentration of free radicals in the cytoplasm and around it.
9. If small molecules of RNA/protein in the ‘counters’ of both centrioles are entirely expended, then such a cell has reached Hayflick limit and the programmed cell death (apoptosis) must be turned on in it. In the non occurrence of the death of such a cell there is a formation of a senescent cell (Fig. 3).

To conclude, it is highly likely that senescent cells emerge because of cellular differentiation during the processes of development and regeneration. Highly differentiated cells (neurons, erythrocytes, hepatocytes and such) always die and are replaced by the new in the regeneration cycles. With each regeneration cycle the amount of senescent cells is increasingly more relative to the reproductive, stem and highly differentiated cells.

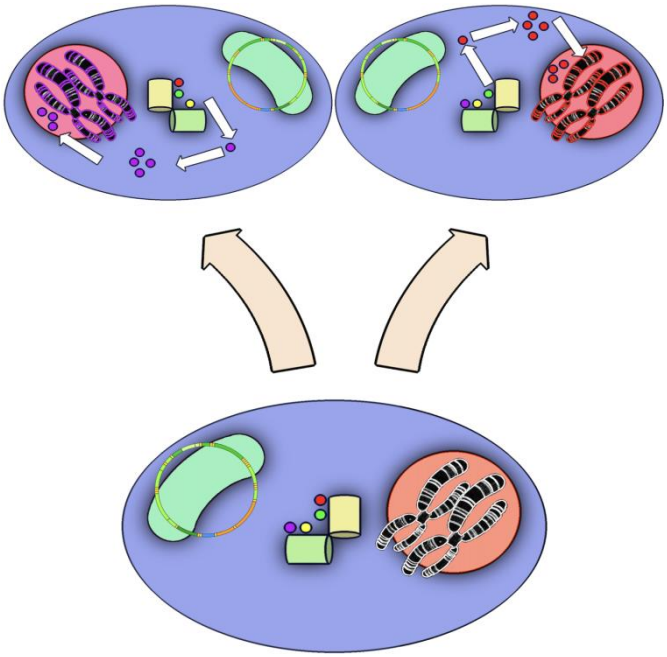

Fig. 2.

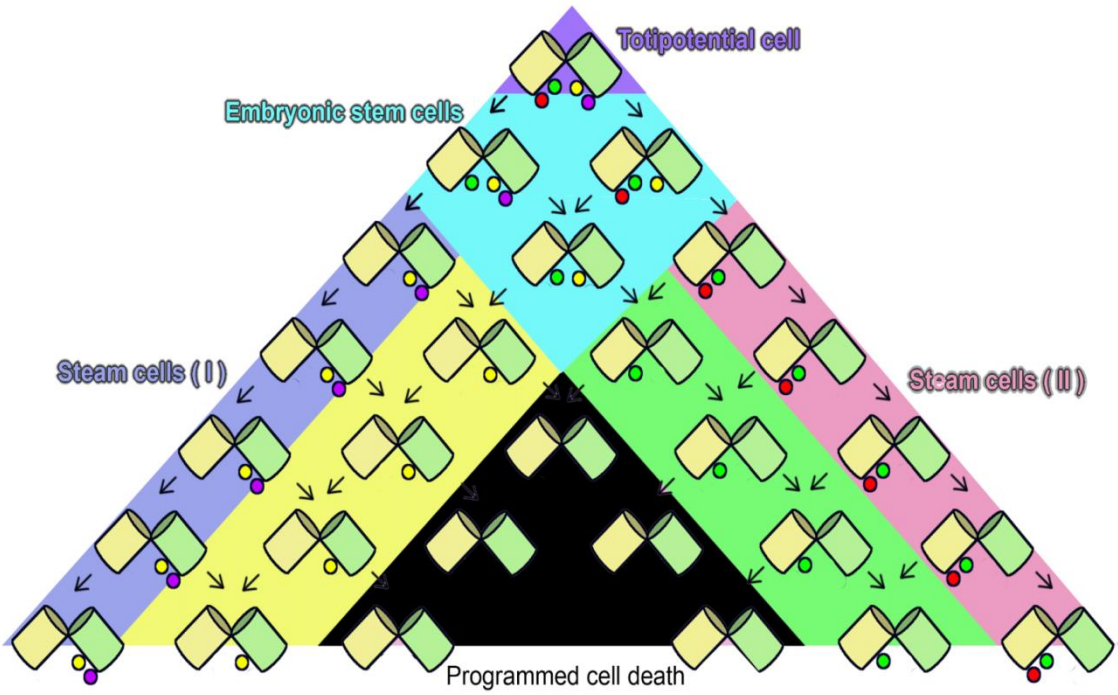

Fig. 3.

To avoid the ageing of the organism due to the growth of intoxication caused by the senescent cells it is necessary to seek methods of periodical partial elimination of highly-toxic senescent cells. The research team from The Mayo Clinic College of Medicine and Science has found an efficient combination of compounds to target the senescent cells in mice, the combination of dasatinib and quercetin [13,14]. The objective of our experiment was to re-conduct the same experiment, but only on humans. As cardiovascular dysfunction is chiefly associated with senescent cells [15], we have made a decision to estimate the outcome by the change of the state of the cardiovascular system.

2. MATERIALS AND METHODS

2.1 Participants

The experiment included middle aged and elderly men between the ages of 36 and 60. Average age was 53,1 years. The majority of them led sedentary lifestyle, others were slightly active, some were smokers, some moderately drank alcohol. The participants did not have any specific illnesses that potentially could affect the experiment. These 64 participants were equally divided into 4 different groups. Such as D+Q, D+P, Q+P and P+P: D+Q group orally administered 50mg dasatinib with 500mg quercetin, D+P group orally administered dasatinib along with placebo, Q+P group took 500mg quercetin with 50mg placebo, P+P group orally administered two compounds of placebo with dosage of 500mg and 50mg accordingly. All participants provided informed consent. The whole research was conducted in accordance with the Helsinki Declaration.

2.2 Reagents

Dasatinib is a inhibitor of multiple tyrosine kinases, used for treating cancers [16] and is known to interfere with EFNB-dependent suppression of apoptosis. Dasatinib (50mg compound) was obtained from LC Laboratories.

Quercetin is a natural flavonol, inhibits PI3K, other kinases, and serpines [17-18]. Quercetin (500 mg compound) was obtained from Sigma-Aldrich Co. (St. Louis, MO, USA; catalog number Q4951).

2.3 Statistical Analysis

All data were presented as mean ± standard deviation. The values between groups for the dependent variables were analyzed to determine if the distributions were normal using Kolmogorov-Smirnov (K-S) Normality test.

Comparison between pre- and post-exercise was performed through the Paired-Samples T test procedure in order to examine the differences between tests. It was also used for comparing the differences for pre-test ME and EE and post-tests ME and EE. Significance level was set at $\alpha = 0.05$ and 0.01. The statistical package program was used to evaluate the results of the study.

2.4 Medical Screening

A medical screening consisting of complete blood count (CBC), C-reactive protein and creatinine blood test were performed on participants a day prior and 21 days after the administration of compounds. CBC covered all basic parameters. Venous blood from participants was collected into anticoagulant EDTA tubes and differential full blood counts were determined on a Beckman Coulter. C-reactive protein concentrations and creatinine blood test were determined on a Beckman auto-analyser.

2.5 Stair ascending test (SAT)

For the accurate assessment of the senolytic effect of all compounds, a stair ascending test [19] was done by participants a day prior to the start of trial and 21 days after the end of trial along with medical screening each time. SAT was performed at a randomly-selected block of flats. Participants were required to ascend 4 floors.

The total vertical distance covered in climbing the 4 floors was 12.48 metres. This vertical distance covered was based on the number of steps climbed and summed up as follows:

Height of a step = 13 cm
Number of steps per flight: 12 steps
Number of flights= 8.
Therefore, vertical height of 4 floors= ((13 cm x 12 steps) x 8 flights)= 1248 cm = 12.48 m.

Participants were clearly acquainted with SAT requirements in advance. They needed to ascend at rapid, but even pace during the whole test and also to take only one step at a time.

They went through a standardized warm-up protocol that consisted of five minutes of stretching for the quadriceps, hamstrings and calf muscles.

Exercise attire was mandatory. Namely, shorts, t-shirts and sports shoes. A five-minute rest interval was initiated between the standardized warm-up and the SAT.

Before the warm-up the systolic (SBP) and diastolic blood pressures (DBP) of participants were measured once. Then right away after reaching 4 floors we measured their SBP and DBP 4 more times with 10 minutes of interval between each.

Overall the whole process took about 40-45 minutes.

2.6 Study in Animals

Researchers from Mayo Clinic tested whether treating 24-month-old mice with D+Q would improve cardiac ejection fraction (the fraction of heart volume pumped during each heart contraction) and vascular responses to acetylcholine, nitroprusside, or U46619 . They gave a single dose of the drugs

and waited 5 days before assaying cardiac function. D and Q are cleared within 48 hours of the last dose.

Despite the fact that mice are relatively resistant to the development of age-related systolic dysfunction, treatment of 24-month-old mice with a single dose of D+Q significantly improved left ventricular ejection fraction and fractional shortening, effects that were mediated by reductions in end-systolic cardiac dimensions but not cardiac preload or alterations in cardiac mass...

Collectively, these data suggest that senescent cells likely exert deleterious effects on cardiovascular function with chronological aging and that acute clearance of senescent cells may be a novel therapeutic approach to improve cardiovascular function and reduce morbidity and mortality from cardiovascular disease in the elderly [20].

3. RESULTS

Results of CBC, C-reactive protein and creatinine blood test performed on participants practically did not show any significant alterations. First SAT results taken a day prior to the administration of compounds showed that the SBP and DBP results of all participants were in the same level with just minor differences. Presented Fig. 4 diagram demonstrates those results measured before the stair ascending test.

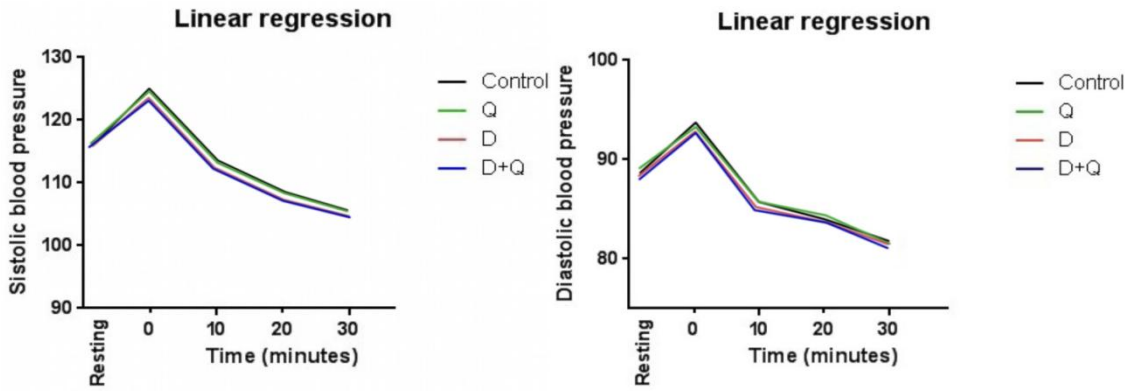

Fig. 4.

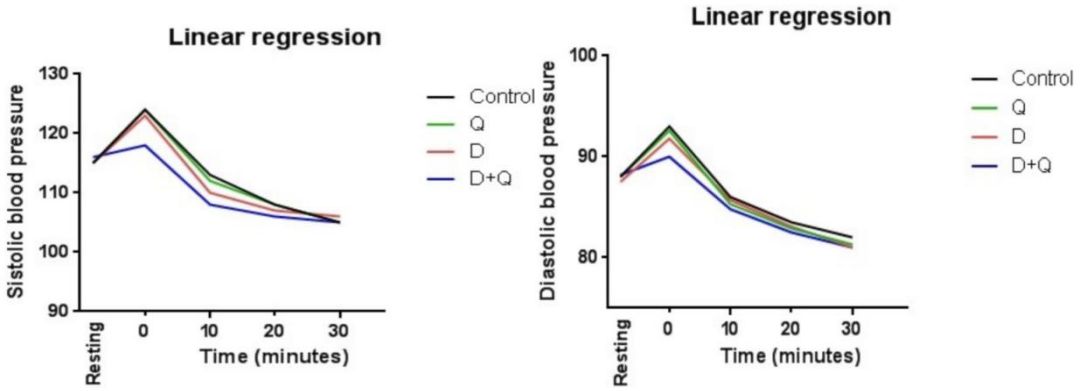

Fig. 5.

Second SAT taken on the 21st day, showcased clear reduction of SBP in groups D+Q and just slight unchanged results in the rest of 2 groups that administered quercetin along with placebo and placebo along with placebo. Significant reduction of SBP was seen starting at 20 minutes and 10 minutes post-exercise respectively in the D+Q group ($p < 0.05$) (Fig. 5), versus the rest of the groups.

As for the DBP it practically remained unaltered.

4. DISCUSSION

We expected a reduction of senescent cells' amount in the organism. In our opinion it might cause a decrease of daily intoxication of tissues, organs and body as a whole. In the case of man it primarily might have an impact on the state of the cardiovascular system.

Results of SAT demonstrated that the human body easily adjusts to strenuous activity and blood pressure promptly comes to its norm. Unfortunately, there is not enough research made on how the return of systolic pressure to the state of rest after exercise changes with age. In the other study [21] similar to ours, 14 IPF diseased patients were given dasatinib and quercetin in other dosage (D:100 mg/day, Q:1250 mg/day, three-days/week over three-weeks), in common with our trial they examined the effect of compounds with the help of strenuous activity. However it consisted of such activities as 6-min walk distance, 4-m gait speed.

It has to be noted, this research also provides initial evidence that senolytics may alleviate physical dysfunction (in their case, not with practically healthy people, but with patients in IPF). Our and this trial warrant evaluation of D+Q in larger randomized controlled trials for senescence-related diseases. Scientists of the other trial reported 16 complications overall, one of which was severe, the rest was either mild or moderate cases. It is highly probable that IPF was not the cause of moderate and particularly more serious complications.

As for the trial conducted by us we did not detect any complications. We attribute this to the fact that minimal (but sufficient) doses were selected. However the fact that the participants of our experiment were practically healthy should also be taken into consideration.

Practically everybody from the D+Q group noted the lightness in the joints the very next day after the first intake of D+Q. In this regard It would be interesting to check the joints before and after in subsequent studies.

But we were not interested in the effects that would have manifested during the presence of D+Q in the body, but the effects that would have manifested after D+Q left the body. It explains why we carried out the second SAT in 16 days after the last administration of D+Q.
The fact that the senolytic effect has remained for so long is explained by realization of partial elimination of ageing cells.

We are inclined to explain the insignificant improvement of D+P group (relative to D+Q combination) by the senolytic effect of dasatinib and non-effect of placebo. The rest controlled Q+P and P+P groups did not show improved results after SAT.

We expected some enhancement in the Q+P group. However it didn't occur in contrast to D+P. Perhaps in the case of Q+P there was a senolytic (small or large with respect to D+Q) effect. But we think that SAT does not detect it. It would be appreciated if the following studies take into account our mistake and make tests that could detect the senolytic effect in the Q+P groups. We believe that for the study of such important combination as D+Q (they do not only enhance physical capacity, but also their combination prevents uterine age-related dysfunction and fibrosis [22] and alleviates intestinal senescence]and inflammation and modulates the gut microbiome in mice [23]), ultimate biochemical standards should be included, at least complete mass-spectrometric analysis of all blood substances.

5. CONCLUSIONS

First group participants who orally administered once a day after meal 50mg of dasatinib and 500mg of quercetin during five days demonstrated remarkably outstanding improvement of physical endurance. They showcased obvious senolytic effect, and their such a simultaneous usage in the

very near future is most likely to be harmless. Senescent cells obviously are involved in the ageing process. It's becoming more clear why people normally age at the same rate. Intensity of ageing and the amount of senescent cells are directly related. The more divisions of cells, the more senescent cells.

COMPETING INTERESTS

Author has declared that no competing interests exist.

REFERENCES

1. Coppé et al. Senescence-associated secretory phenotypes reveal cell-nonautonomous functions of oncogenic RAS and the p53 tumor suppressor. PLoS Biol; 2008.
   DOI: 10.1371/journal.pbio.0060301
2. Coppé et al. The senescence-associated secretory phenotype: the dark side of tumor suppression. Annu Rev Pathol; 2010.
   DOI: 10.1146/annurev-pathol-121808-102144.
3. Tkemaladze J, Chichinadze K. Centriolar mechanisms of differentiation and replicative aging of higher animal cells. Biochemistry (Mosc); 2005a.
   DOI: 10.1007/s10541-005-0261-6
4. Tkemaladze J, Chichinadze K. Centriole, differentiation, and senescence. Rejuvenation Res; 2010.
   DOI: 10.1089/rej.2009.0904
5. Hartung, Stahl. Preleptotene chromosome condensation in mouse oogenesis. Cytogenet Cell Genet; 1977.
   DOI: 10.1159/000130777
6. Abumuslimov et al. An electron microscopic study of centriole and centrosome morphogenesis in the early development of the mouse. Tsitologiia; 1994.
7. Maro et al. Cell polarity and microtubule organisation during mouse early embryogenesis. Dev Suppl; 1191.
8. Ishiuchi, Torres-Padilla. Towards an understanding of the regulatory mechanisms of totipotency. Curr Opin Genet Dev; 2013.
   DOI: 10.1016/j.gde.2013.06.006.
9. Chichinadze K, Tkemaladze D. [Centrosomal hypothesis of cellular aging and differentiation]. Adv Gerontol; 2008.
   PMID: 19432168.
10. Roy, Kundu. Gene regulatory networks and epigenetic modifications in cell differentiation. IUBMB Life; 2014.
    DOI: 10.1002/iub.1249
11. Hayflick L. Mortality and immortality at the cellular level. A review. Biochemistry (Mosc); 1997.
12. Tkemaladze J, Chichinadze K. Potential role of centrioles in determining the morphogenetic status of animal somatic cells. Cell Biol Int; 2005b.
    DOI: 10.1016/j.cellbi.2005.03.003. PMID: 15886028.
13. Roos et al. Chronic senolytic treatment alleviates established vasomotor dysfunction in aged or atherosclerotic mice. Aging Cell; 2016.
    DOI: 10.1111/acel.12458.
14. Campisi, d'Adda di Fagagna. Cellular senescence: when bad things happen to good cells. Nat Rev Mol Cell Biol; 2007.
15. Kirkland J, Tchkonia. Clinical strategies and animal models for developing senolytic agents. Exp Gerontol; 2014.
    DOI: 10.1016/j.exger.2014.10.012.
16. Montero et al. Inhibition of SRC family kinases and receptor tyrosine kinases by dasatinib: possible combinations in solid tumors. Clin Cancer Res; 2011.
    DOI: 10.1158/1078-0432.CCR-10-2616.
17. Bruning, 2013. Inhibition of mTOR signaling by quercetin in cancer treatment and prevention. Anticancer Agents Med Chem.

18. Olave et al., Upstream stimulatory factor-2 mediates quercetin-induced suppression of PAI-1 gene expression in human endothelial cells. J Cell Biochem
DOI: 10.1002/jcb.22760.
19. The, Aziz. A stair-climb test of cardiorespiratory fitness for Singapore. Singapore Med J; 2000.
PMID: 11296784
20. Zhu et al. The Achilles' heel of senescent cells: from transcriptome to senolytic drugs. Aging Cell; 2015.
DOI: 10.1111/acel.12344.
21. Justice et al. Senolytics in idiopathic pulmonary fibrosis: Results from a first-in-human, open-label, pilot study. EBioMedicine; 2019.
DOI: 10.1016/j.ebiom.2018.12.052.
22. Cavalcante MB, Saccon TD, Nunes AD, Kirkland JL, Tchkonia T, Schneider A, Masternak MM. Dasatinib plus quercetin prevents uterine age-related dysfunction and fibrosis in mice. Aging (Albany NY). 2020;12(3):2711.
23. Saccon TD, Nagpal R, Yadav H, Cavalcante MB, Nunes AD, Schneider A, Gesing A, Hughes B, Yousefzadeh M, Tchkonia T, Kirkland JL. Senolytic combination of Dasatinib and Quercetin alleviates intestinal senescence and inflammation and modulates the gut microbiome in aged mice. The Journals of Gerontology: Series A; 2021.

Biography of author(s)

Tkemaladze Jaba
International Longevity Centre- 0131, King Demetre Tavdadebuli 81, Tbilisi, Georgia.

Research and Academic Experience:
International Longevity Centre - Georgia
February 2021 - Present
Department of Genetics
Head of Department

International Longevity Centre - Georgia
January 2016 - January 2021
Department of Genetics
Scientific Researcher

Georgian State Public Broadcaster
September 2008 - August 2009
Medical blog of "Alioni" programme
Anchor

Ivane Javakhishvili Tbilisi State University
October 1999 - December 2015
Aleksandre Natishvili Institute of Morphology
Scientific Researcher

Research Area: His Research Area includes Molecular Biology, General Medicine, and Genetics.

Number of Published papers: He has 13 Research publication.

Any other remarkable point(s): He is an Author of The Cytogenetic Centriolar Theory of Ageing.

© Copyright (2022): Author(s). The licensee is the publisher (B P International).

DISCLAIMER
This chapter is an extended version of the article published by the same author(s) in the following journal.
Journal of Biomedical Sciences, 8(3:3): 2019.